Instrument Validity as an Audit for Algorithmic Collective Action

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Abstract

We study encouragement experiments in which a group coordinates its treatment take-up. A one-sided population discrepancy $V(P)$ is the sharp lower bound on the share of defiers. The bound is zero exactly when the defiers' potential-outcome measures are dominated by the compliers', so detectability depends on outcome distributions as well as group size. In a discretized Gaussian simulation, the complier-LATE set widens with the defier share under outcome mimicry but is non-monotone under separation. With a 29% defier share, the Wald estimand is -0.45 when the complier effect is 1.0; $V(P) = 0.0022$ and a moment-inequality test rejects in 18% of $n = 200{,}000$ replications. The first stage is 0.01 in the same design. We therefore report the monotonicity inequalities together with instrument strength.

1 Introduction

Algorithmic collective action studies how coordinated users can change a platform's learned model or recommendations. Existing work considers pooled changes to training data [Hardt et al., 2023], playlist reordering [Baumann and Mendler-Dünner, 2024], and interactions among several collectives [Karan et al., 2025, Battiloro et al., 2025].

We ask whether a randomized platform experiment can audit such coordination without observing group membership or communication.

We consider coordination within a randomized platform experiment. Instead of modifying features or training examples, group members invert the assigned encouragement. In the principal-strata terminology, they behave as defiers. The standard local average treatment effect framework rules out this response under its monotonicity assumption [Imbens and Angrist, 1994, Angrist et al., 1996].

Monotonicity and the other LATE assumptions imply inequalities in observed outcome distributions [Balke and Pearl, 1997, Kitagawa, 2015, Mourifié and Wan, 2017]. We show that the population magnitude of their violation is also a sharp lower bound on the defier share. Under a recruitment model, this share is the fraction of active group members.

Contributions.

- Theorem 1 proves that $V(P)$ is the sharp lower bound on defier mass. A linear-program check over 30 discretized DGPs agrees to within 8.3×10^{-8}.
- Proposition 2 characterizes when $V(P) = 0$: the defiers' potential-outcome measures are dominated by the corresponding complier measures.
- Finding 1 shows numerically that the complier-LATE set need not widen monotonically with the defier share. It widens under mimicry but can narrow when outcomes are well separated.

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- The simulations compare the audit with a first-stage diagnostic. Each detects failures that the other can miss.

Related work. Kowalski [2019] and Christy and Kowalski [2024] use design-based likelihoods to learn about defiers from experimental sample composition. de Chaisemartin [2017], Dahl et al. [2023], and Kédagni et al. [2024] derive treatment-effect results under alternatives to monotonicity. Our object is a closed-form positive-part functional on general outcome spaces. Interpreting defier mass as coalition size requires the recruitment model below. Related work on partial identification with LLM reports treats the model as a noisy measurement device rather than as a noncompliant unit [Chen et al., 2026].

2 Setup

A platform randomizes an encouragement $Z \in \{0,1\}$, observes a binary take-up decision $D \in \{0,1\}$ and an outcome Y in a Polish space $\mathcal{Y}$. Potential take-up is $D(z)$ and potential outcomes are $Y(z,d)$. Principal strata are $T = (D(0), D(1))$, with $a = (1,1)$ always-takers, $n = (0,0)$ never-takers, $c = (0,1)$ compliers and $f = (1,0)$ defiers. We maintain throughout

Assumption 1 (Random assignment)**.** $Z \perp\!\!\!\perp \big(Y(\cdot,\cdot), D(\cdot)\big)$.

Assumption 2 (Exclusion)**.** $Y(z,d) = Y(d)$ *almost surely.*

We do *not* impose monotonicity, $\mathbb{P}(T = f) = 0$. Under random assignment, monotonicity is the restriction that yields the usual threshold-crossing representation for binary take-up [Vytlacil, 2002].

The collective. In the baseline recruitment model, an active share α inverts the platform's encouragement and all non-members satisfy monotonicity, so $\mathbb{P}(T = f) = \alpha$. The results always bound defier mass. With incomplete participation they bound the active coordinating share, not nominal membership.

Observables. Write $\mu_t^{(d)}(B) = \mathbb{P}(T{=}t, Y(d) \in B)$ and let $m_{dz}(B) = \mathbb{P}(Y \in B, D{=}d \mid Z{=}z)$ be the four observed sub-distributions. Under Assumptions 1–2,

$$m_{11} = \mu_a^{(1)} + \mu_c^{(1)}, \qquad m_{10} = \mu_a^{(1)} + \mu_f^{(1)}, \qquad m_{00} = \mu_n^{(0)} + \mu_c^{(0)}, \qquad m_{01} = \mu_n^{(0)} + \mu_f^{(0)}.$$

Always-takers and never-takers appear in both members of each pair and cancel in the differences

$$S_1 := m_{10} - m_{11} = \mu_f^{(1)} - \mu_c^{(1)}, \qquad S_0 := m_{01} - m_{00} = \mu_f^{(0)} - \mu_c^{(0)}, \tag{1}$$

so the two observable signed measures S_1, S_0 carry the defiers. Let $S = S_+ - S_-$ denote the Jordan decomposition and define the **violation functional**

$$V(P) := \max\Big\{ \sup_B S_1(B),\ \sup_B S_0(B) \Big\} = \max\big\{ S_{1+}(\mathcal{Y}),\ S_{0+}(\mathcal{Y}) \big\}. \tag{2}$$

Under monotonicity $S_1 = -\mu_c^{(1)} \le 0$ and $S_0 = -\mu_c^{(0)} \le 0$, recovering the Balke–Pearl/Kitagawa inequalities and $V(P) = 0$ [Balke and Pearl, 1997, Kitagawa, 2015]. The functional is a one-sided set-indexed discrepancy between sub-probability measures.

3 The audit

Theorem 1 ($V(P)$ is the sharp lower bound on defier mass)**.** *Let P be any distribution of (Z, D, Y) generated by some model satisfying Assumptions 1–2. Then*

$$\alpha = \mathbb{P}(T = f) \ \ge\ V(P),$$

and the bound is sharp: there exists a model satisfying Assumptions 1–2, inducing exactly P, with $\mathbb{P}(T = f) = V(P)$.

The lower bound follows from (1) because $S_1(B) \le \mu_f^{(1)}(B) \le \mathbb{P}(T = f)$ for every B. For attainment, the construction sets $\mu_f^{(1)} = S_{1+}$ and $\mu_c^{(1)} = S_{1-}$ on the $d = 1$ side, uses the symmetric

construction on the $d = 0$ side, and adds mass to the slack side until both defier measures have mass $V(P)$. Appendix A gives the construction and verifies feasibility.

The functional uses the standard monotonicity inequalities and provides a lower bound, not an upper bound, on defier mass. Its plug-in estimator has positive-part bias. We therefore report it as an effect size and test the finite cell inequalities with a separately studentized, least-favourable Gaussian critical value (Section 5).

4 Limits of the audit

Proposition 2 (When violations are undetectable)**.** *$V(P) = 0$ if and only if $\mu_f^{(1)} \le \mu_c^{(1)}$ and $\mu_f^{(0)} \le \mu_c^{(0)}$ as measures. In particular, a collective whose members' potential-outcome distributions coincide with the compliers' is exactly invisible to the audit whenever $\alpha \le \mathbb{P}(T = c)$, for any α.*

This follows from (2): $S_{1+}(\mathcal{Y}) = 0$ if and only if $S_1 \le 0$. The audit does not distinguish coordination from other sources of defiance. It also has no power against a group drawn from the complier population whose outcome behaviour is unchanged and whose only change is take-up under the encouragement.

The effect on the platform's reported estimate can nevertheless be large. Let τ_c and τ_f be the average treatment effects of compliers and group members.

Proposition 3 (Influence)**.** *If $\alpha = \mathbb{P}(T = f) < \mathbb{P}(T = c)$, the platform's Wald estimand satisfies* $\mathrm{Wald} = \big(\mathbb{P}(T{=}c)\tau_c - \alpha\tau_f\big)/\big(\mathbb{P}(T{=}c) - \alpha\big)$*, so its bias relative to the complier LATE is*

$$\big|\mathrm{Wald} - \tau_c\big| \;=\; \frac{\alpha\,\big|\tau_c - \tau_f\big|}{\mathbb{P}(T{=}c) - \alpha},$$

which diverges as $\alpha \uparrow \mathbb{P}(T = c)$.

Indeed, exclusion makes the reduced-form effect $\mathbb{P}(T = c)\tau_c - \alpha\tau_f$ and the first stage $\mathbb{P}(T = c) - \alpha$; division and subtraction of τ_c give the formula. Thus influence requires effect heterogeneity, $\tau_f \neq \tau_c$. Taking $\alpha = 0.29$ against $\mathbb{P}(T = c) = 0.30$ with $\tau_f = 1.05$ and $\tau_c = 1$, the platform reports $\mathrm{Wald} = -0.45$ while $V(P) = 0.0022$.

We next examine how defiers affect identification of the complier effect.

Finding 1 (Information loss in the discretized simulation). The width of the sharp complier-LATE set for the discretized outcome model is *not* monotone in α in general. An exact linear-fractional program (Figure 1c) gives an increase from 1.30 to 6.30 as α moves from 0 to 0.29 under perfect mimicry; for separation 3, the width is non-monotone and moves from 1.30 to 1.31.

When defier outcomes are well separated, the linear program can allocate much of their outcome mass to the defier stratum. Identification can then improve even as the defier share grows. Under mimicry, that allocation remains ambiguous.

Finding 2 (Detectability and information loss in this family). Within the simulated Gaussian family, lower outcome separation produces smaller $V(P)$ and a wider complier-LATE set at large α. This numerical relationship is not asserted as a universal ordering across collective strategies.

5 Experiments

We simulate a platform experiment with a planted collective. Population quantities are exact; detection rates use 100 replications. We test the 24 cellwise monotonicity inequalities with a studentized maximum statistic and 1000 draws from its least-favourable plug-in Gaussian law at level 0.05. Separately, we estimate the effect size $V(P)$ over unions of 12 histogram cells. The restricted class understates the population supremum, but finite-sample positive-part bias means the plug-in value is not itself a confidence bound.

Separation can be detected. Under full outcome separation, $\alpha = 0.01$ gives $V(P) = 0.005$ and rejection rates 0.08, 0.98, and 1.00 at $n = 2{,}000$, $10{,}000$, and $50{,}000$. At $n = 2{,}000$, the rates rise to 0.63 for $\alpha = 0.02$ and 1.00 for $\alpha = 0.05$. These rates are specific to the Gaussian cells and test implementation.

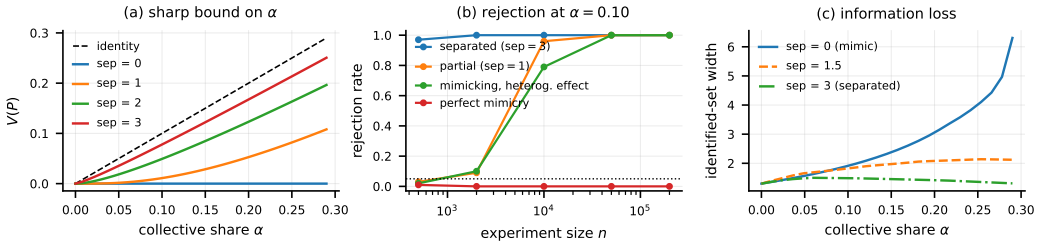


Figure 1: (a) The sharp bound $V(P)$ against defier share α at four outcome separations. (b) Rejection rates of the studentized Gaussian moment-inequality test at $\alpha = 0.10$; exact mimicry lies inside the composite null, so rejection can be below 0.05. (c) Exact complier-LATE set width for the discretized model (Finding 1).

Exact mimicry lies inside the null. At $\alpha = 0.10$ (Figure 1b), separation 3 is detected with rate 0.97 at $n = 500$; separation 1 reaches 0.96 at $n = 10{,}000$. Under exact mimicry, rejection rates are 0.01 at $n = 500$ and 0.00 at all larger sizes tested. This is compatible with a level-0.05 composite-null test: the mimicking DGP satisfies the inequalities strictly, so rejection need not equal the nominal level.

The first stage is complementary. The sign-flipping configuration has only 18% rejection at $n = 200{,}000$ and a first stage of 0.01, which signals a weak instrument. At $\alpha = 0.05$ with $\tau_f = 3$, however, the Wald estimand is 0.60 against a complier effect of 1.0, with $V(P) = 0.018$ and first stage 0.25. Reporting both diagnostics reveals different weaknesses but does not by itself identify the complier effect.

6 Discussion

Interpreting defier mass as an active coordinating share requires the recruitment model. The population identification results are exact, but the reported rejection rates depend on Gaussian outcomes, a fixed partition, and an asymptotic test. We also do not model strategic interaction between the group and the auditor. Appendix B discusses these limitations, and Appendix C discusses the dual-use implication of Proposition 2.

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A Proof of Theorem 1

Fix P in the image of the model, i.e. generated by some $\mu^\star = (\mu_a^{(1)}, \mu_n^{(0)}, \mu_c^{(1)}, \mu_c^{(0)}, \mu_f^{(1)}, \mu_f^{(0)})$ satisfying Assumptions 1–2.

Lower bound. By (1), for any Borel B, $S_1(B) = \mu_f^{(1)}(B) - \mu_c^{(1)}(B) \le \mu_f^{(1)}(B) \le \mathbb{P}(T = f)$, so $S_{1+}(\mathcal{Y}) \le \mathbb{P}(T = f)$. The same argument applies to S_0, hence $V(P) \le \mathbb{P}(T = f) = \alpha$ in the baseline recruitment model.

Attainment. Let $\varphi := V(P)$ and recall $\delta = \mathbb{P}(T = f) - \mathbb{P}(T = c) = m_{10}(\mathcal{Y}) - m_{11}(\mathcal{Y}) = m_{01}(\mathcal{Y}) - m_{00}(\mathcal{Y})$ is identified. We construct a model inducing P with $\mathbb{P}(T = f) = \varphi$.

Any candidate must satisfy $\mu_f^{(1)} - \mu_c^{(1)} = S_1$ with both measures nonnegative, whence $\mu_f^{(1)} \ge S_{1+}$; the minimal choice is $\mu_f^{(1)} = S_{1+}$, $\mu_c^{(1)} = S_{1-}$. Since S_{1+} is carried by the set where $S_1 > 0$ and there $S_{1+} = m_{10} - m_{11} \le m_{10}$, the induced $\mu_a^{(1)} := m_{10} - S_{1+} = m_{11} - S_{1-}$ is a nonnegative measure; the two expressions agree because $m_{10} - m_{11} = S_1 = S_{1+} - S_{1-}$. Define $S_{0\pm}$, $\mu_n^{(0)}$ analogously.

To raise both sides to the common mass φ, put $\mu_f^{(1)} = S_{1+} + R_1$, $\mu_c^{(1)} = S_{1-} + R_1$ with $R_1 \ge 0$, $R_1(\mathcal{Y}) = \varphi - S_{1+}(\mathcal{Y})$, and likewise R_0 with mass $\varphi - S_{0+}(\mathcal{Y})$. Both differences in (1) are preserved for any such R_1, R_0, and nonnegativity of $\mu_a^{(1)} = m_{10} - S_{1+} - R_1$ requires only $R_1 \le \mu_a^{(1)}$, so a valid R_1 exists iff

$$\varphi - S_{1+}(\mathcal{Y}) \;\le\; \mu_a^{(1)}(\mathcal{Y}) = m_{10}(\mathcal{Y}) - S_{1+}(\mathcal{Y}) \quad \Longleftrightarrow \quad \varphi \le m_{10}(\mathcal{Y}) = \mathbb{P}(D{=}1 \mid Z{=}0),$$

that is $\varphi \le \mathbb{P}(D{=}1 \mid Z{=}0)$, and symmetrically $\varphi \le m_{01}(\mathcal{Y}) = \mathbb{P}(D{=}0 \mid Z{=}1)$. Both hold: defiers have $D(0) = 1$ and $D(1) = 0$, so for the true model $\mathbb{P}(T = f) \le \min\{\mathbb{P}(D{=}1 \mid Z{=}0), \mathbb{P}(D{=}0 \mid$

$Z{=}1)\}$, and $\varphi = V(P) \leq \mathbb{P}(T = f)$ by the first part. (This step uses only that *some* valid model generates P, so the argument is not circular.)

It remains to check the type shares sum to one. With $\mathbb{P}(T = f) = \varphi$, $\mathbb{P}(T = c) = \varphi - \delta$, $\mathbb{P}(T = a) = m_{10}(\mathcal{Y}) - \varphi$ and $\mathbb{P}(T = n) = m_{01}(\mathcal{Y}) - \varphi$, the total is $m_{10}(\mathcal{Y}) + m_{01}(\mathcal{Y}) - \delta = m_{01}(\mathcal{Y}) + m_{11}(\mathcal{Y}) = \mathbb{P}(D{=}0 \mid Z{=}1) + \mathbb{P}(D{=}1 \mid Z{=}1) = 1$. Marginal masses match across d by construction ($\mu_c^{(1)}(\mathcal{Y}) = \mu_c^{(0)}(\mathcal{Y}) = \varphi - \delta$ and $\mu_f^{(1)}(\mathcal{Y}) = \mu_f^{(0)}(\mathcal{Y}) = \varphi$), and a joint law for $(Y(0), Y(1))$ within each stratum is obtained by any coupling of the stratum marginals, e.g. the independent one. Hence the constructed model satisfies Assumptions 1–2, induces P, and has $\mathbb{P}(T = f) = V(P)$. □

Numerical check. Independently of the proof, we minimized $\mathbb{P}(T = f)$ by linear programming over the discretized feasible set (four equalities per outcome cell, equal stratum mass across d, total mass one, nonnegativity) and compared with the cellwise $V(P)$ across 30 data-generating processes spanning $\alpha \in \{0.02, \ldots, 0.29\}$, separations $\{0, 1, 2.5\}$ and $\tau_f \in \{1, 2\}$. The maximum absolute discrepancy was 8.3×10^{-8}, at linear-program tolerance.

B Limitations in detail

Defiance versus coordination. Theorem 1 bounds defier mass. Reading it as active coordination requires the recruitment model: active members invert encouragement and non-members do not. Psychological reactance, misread instructions, or interface confusion can also produce defiers; without ruling these out, $V(P)$ is not a coalition-size lower bound. With incomplete collective participation but no other defiers, it remains a lower bound on active and nominal membership. Distinguishing coordinated from idiosyncratic defiance requires auxiliary timing or network information.

Model specificity of rejection rates. Theorem 1 and Propositions 2–3 are population results for the general model. Rejection rates use Gaussian outcomes, 12 fixed cells, and a least-favourable Gaussian approximation. Exact mimicry is inside the composite null, so rejection can be below the nominal level; outcome-separated alternatives can be detected quickly in this parameterization. Different outcome laws, partitions, and moment-selection refinements will change the rates.

No equilibrium analysis. We compute detectability/influence pairs in one parametric family (Appendix D) but do not solve for equilibrium. A full treatment would specify the collective's objective, the platform's audit policy and false-positive cost, and the timing of play. Finding 2 is numerical and does not establish a general ordering of strategies.

C Broader impact of the stealth characterization

Proposition 2 characterizes when an identification-based audit has no population violation and is therefore dual-use. Publishing the condition helps auditors avoid interpreting non-rejection as evidence of no coordination. The paired first-stage diagnostic reveals weak-instrument cases but is not a complete mitigation, as the healthy-first-stage example shows. The result concerns aggregate detectability in a stylized model and provides no operational capability against a specific system.

D The stealth frontier

Figure 2 provides the calculation behind Finding 2. It is not a universal feasibility set or an equilibrium solution.

E Simulation details

Outcomes are Gaussian with unit variance. Compliers have $Y(0) \sim \mathcal{N}(0, 1)$ and $Y(1) \sim \mathcal{N}(\tau_c, 1)$ with $\tau_c = 1$; defiers are shifted in level by sep and have effect τ_f. Baseline shares are $\mathbb{P}(T = c) = 0.30$ and $\mathbb{P}(T = a) = 0.25$. Outcomes are winsorized into $[-6, 9]$ and discretized into 12

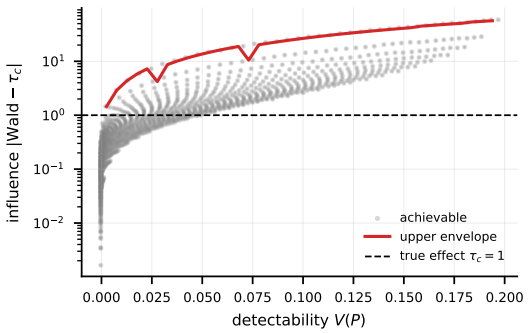


Figure 2: Pairs of $V(P)$ and $|\mathrm{Wald} - \tau_c|$ in the simulated family at $\mathrm{sep} = 0$, sweeping $\alpha \in [0.01, 0.29]$ and $\tau_f \in [1, 3]$. The curve is a numerical envelope for this parameterization, not a general equilibrium frontier.

cells. The complier-LATE ratio is optimized exactly for this discretized model using the Charnes–Cooper transformation; no profiling grid is used. Detection uses 100 replications and 1000 Gaussian calibration draws per replication.

For testing, stack the 24 cell moments $g = (m_{10} - m_{11}, m_{01} - m_{00})$. Their covariance is the sum of the two empirical arm-specific multinomial covariance matrices divided by arm size. The statistic is $\max_j \hat{g}_j / \widehat{\mathrm{se}}_j$; its critical value is the 0.95 quantile of the maximum of a zero-mean Gaussian with the plug-in correlation matrix. Setting all null moments to zero is least favourable, yielding asymptotic size control under the usual nondegeneracy conditions.

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